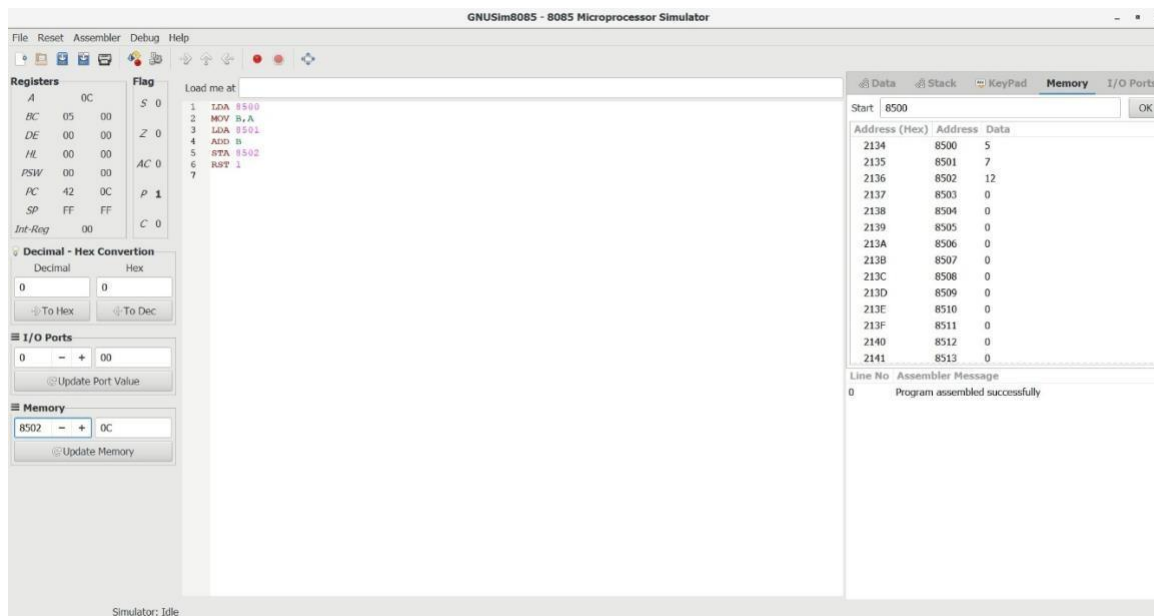


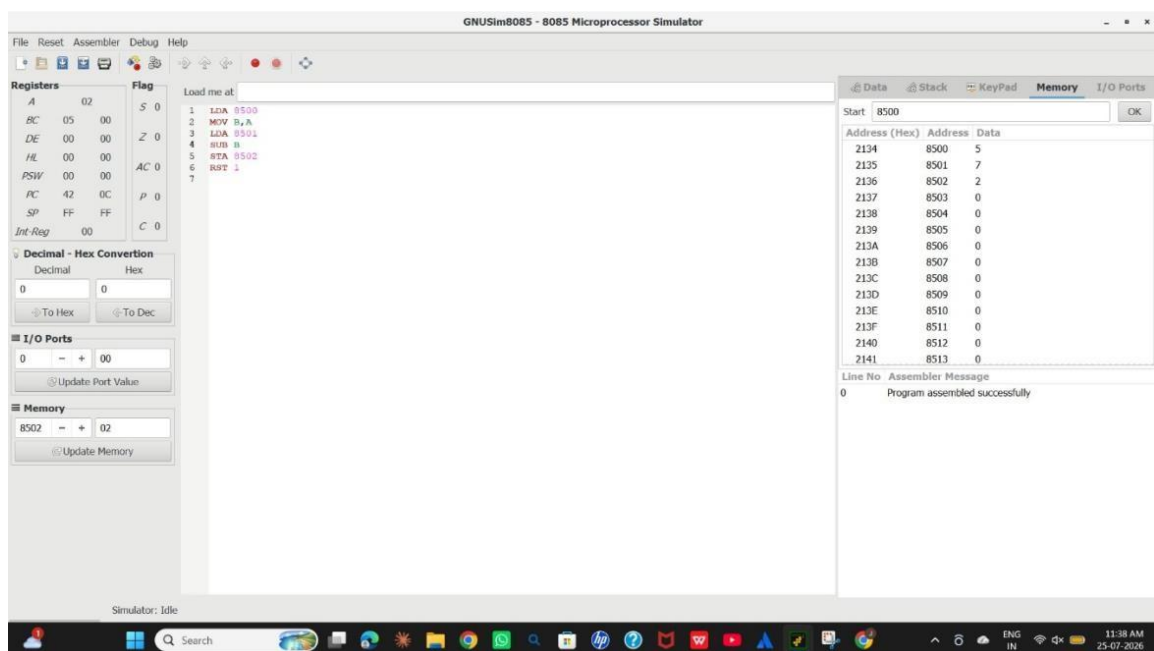
CSA1204 COMPUTER ARCHITECTURE

8085 LAB PROGRAMS

1. Addition of 2 numbers.



2. Subtraction of 2 numbers.



3. Addition with carry.

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 STC
5 ADC B
6 STA 2502
7 HLT
```

The registers on the left show the following values:

Register	Value
A	19
BC	00
DE	00
HL	00
PSW	00
PC	42
SP	FF
Int-Reg	00

The flags on the right show: S=0, Z=0, AC=0, P=0, C=0.

The memory window on the right shows the following data:

Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	35
09C6	2502	61
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message window at the bottom right shows: "Program assembled successfully".

4. Subtraction with borrow.

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 STC
5 SBB B
6 STA 2502
7 HLT
```

The registers on the left show the following values:

Register	Value
A	09
BC	19
DE	00
HL	00
PSW	00
PC	42
SP	FF
Int-Reg	00

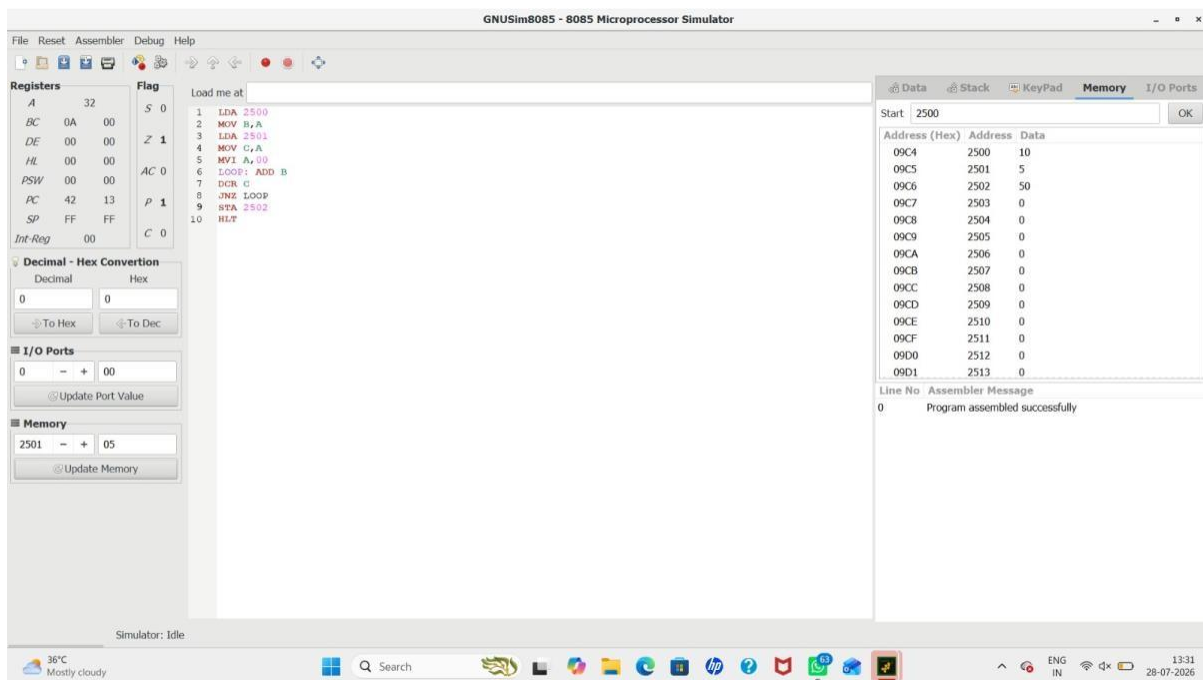
The flags on the right show: S=0, Z=0, AC=0, P=1, C=0.

The memory window on the right shows the following data:

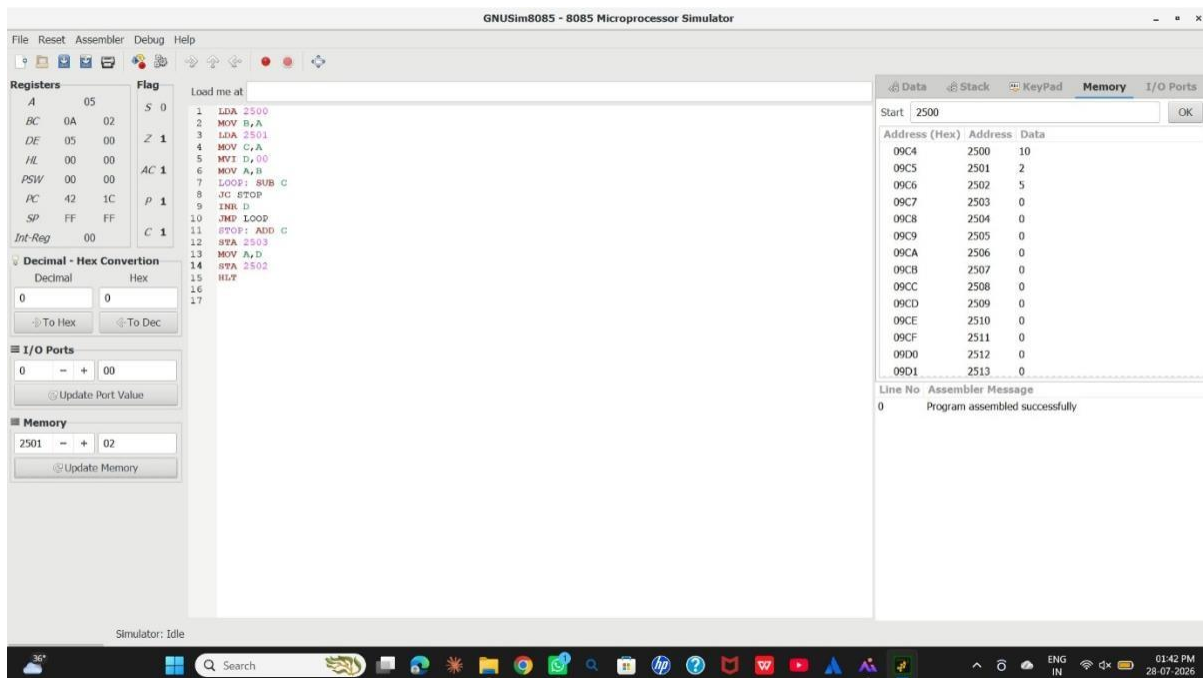
Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	35
09C6	2502	9
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message window at the bottom right shows: "Program assembled successfully".

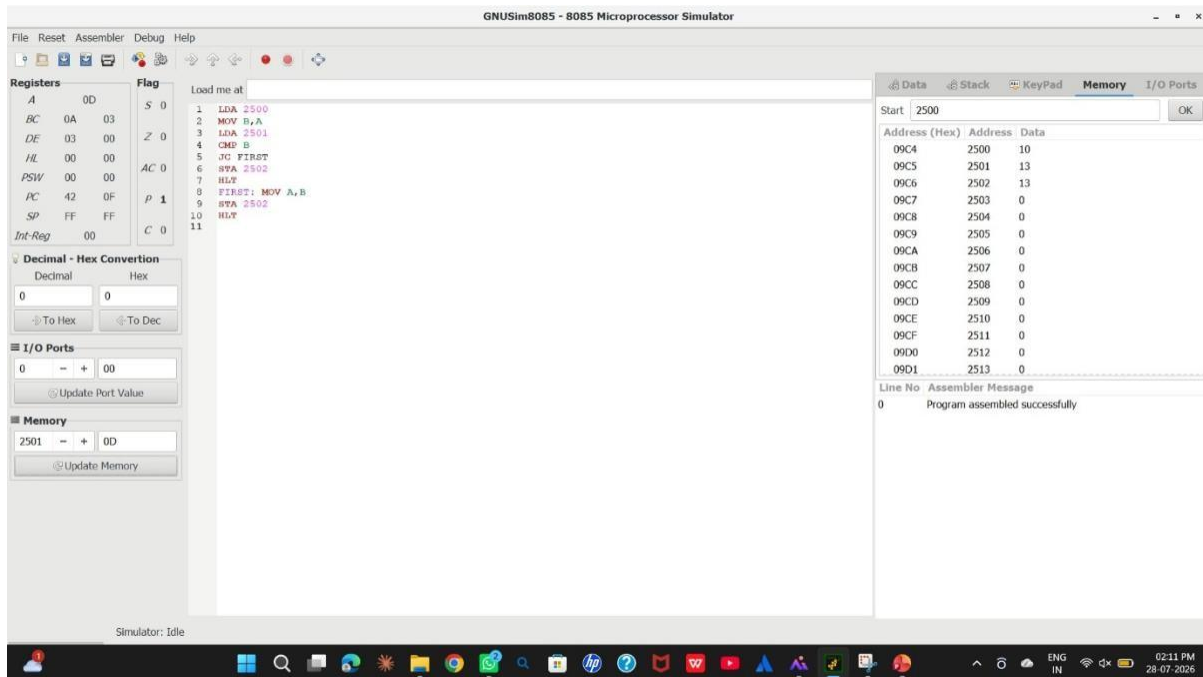
5. Multiplication of 2 numbers.



6. Division of two numbers



7. Greatest of 2 numbers.



8. Smallest of 2 numbers.

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value	Flag
A	0A	S 0
BC	0A 03	Z 0
DE	03 00	AC 0
HL	00 00	P 1
PSW	00 00	C 0
PC	42 10	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 -- + 00

Update Port Value

Memory

2501 -- + 0D

Update Memory

Load me at

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 CMP B
5 JG SMALL
6 MOV A,B
7 SMALL: STA 2502
8 HLT
9
```

Start 2500 OK

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	13
09C6	2502	10
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

0216 PM 28-07-2025